Joseph (Joe) D. Osborn

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https://jdosbo.github.io

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Research Interests

High-energy experimental nuclear physics, software and computing in high-energy physics.

Education

University of Michigan, Ph.D. in Physics, 2018.
University of Michigan, M.S. in Physics, 2015.
University of Kentucky, B.S. in Physics, B.S. in Mathematics, 2013.

Software and Computational Fluency

Languages: C/C++, Java, Python, LaTeX, Bash
Tools: git, Jenkins, Linux, UML, Doxygen, Docker, Singularity, Serverless/FaaS, pybind
HEP Packages: ROOT, ACTS, GEANT, Fun4All, RooUnfold

Ph.D. Thesis

Nonperturbative factorization breaking and color entanglement effects in dihadron and direct photon-hadron angular correlations in p+p and p+A collisions. J. D. Osborn, University of Michigan, May 29, 2018. hep-ex/1806.07763.

Research Positions

Associate Scientist, Brookhaven National Laboratory
09/2025-present
Physics Department

Assistant Scientist, Brookhaven National Laboratory
08/2022-09/2025
Physics Department

Associate Research Scientist, Oak Ridge National Laboratory
06/2021-08/2022
Computing and Computational Sciences Directorate

Postdoctoral Research Associate, Oak Ridge National Laboratory
07/2019-05/2021
Computing and Computational Sciences Directorate

Postdoctoral Research Fellow, University of Michigan
06/2018-06/2019
Department of Physics

Graduate Research Assistant, University of Michigan
07/2013-06/2018
Thesis advisor: Christine Aidala
Department of Physics

Research Funding

1. Laboratory Directed Research and Development-B, Brookhaven National LAboratory. *Global event reconstruction with Foundation Models using real experimental data.* 10/2026-09/2028, $500k.

2. Laboratory Directed Research and Development-C, Brookhaven National Laboratory. *Sparse Data Preparation for Event Reconstruction Foundational Models.* 02/2025-09/2025, $62k.

3. Oak Ridge National Laboratory subcontract through Brookhaven National Laboratory. *Software Development for sPHENIX.* 03/2020-07/2022. PI, $466k.

Peer Reviewed Publications with Significant Contribution

For full publication list search 'find a J. D. Osborn' on inspirehep.net

1. Y. Ren et. al. "A roadmap towards scaling, reasoning, and self-evolving foundation model for nuclear and particle physics." International Journal of Modern Physics A, 41, 2650086 (2026).

2. D. Park et al. "FM4NPP: A Scaling Foundation Model for Nuclear and Particle Physics." arXiv:2508.14087, accepted by ICLR.

3. Y. Huang et. al. "Variable Rate Neural Compression for Sparse Detector Data." Patterns 7, 3 101452 (2026).

4. C. Agapopoulou et. al. (HEP Software Foundation), "The critical importance of software in HEP." Eur. Phys. J. C 85 (2025) 1142.

5. J. K. Adkins et. al. "Design of the ECCE detector for the Electron Ion Collider." Nuclear Instruments and Methods in Physics A, 2025, 170240 (2025).

6. C. Allaire et al. "Artificial Intelligence for the Electron Ion Collider (AI4EIC)". Computing and Software for Big Science 8, 5 (2024).

7. N. J. Abdulameer et. al. (PHENIX Collaboration), "Transverse single-spin asymmetry of midrapidity π^0 and η mesons in p+Au and p+Al collisions at $\sqrt{s_{NN}} = 200$ GeV." Phys. Rev. D 107, 112004 (2023).

8. U. A. Acharya et al. (PHENIX Collaboration), “Improving constraints on gluon spin-momentum correlations in transversely polarized protons via midrapidity open-heavy-flavor electrons in $p^{\uparrow} + p$ collisions at $\sqrt{s} = 200$ GeV.” Phys. Rev. D 107, 052012 (2023).

9. J. C. Bernaeur et al. “Scientific computing plan for the ECCE detector at the Electron Ion Collider” NIMA 1047, 167859 (2023).

10. R. Abdul Khalek et al. “Science Requirements and Detector Concepts for the Electron-Ion Collider: EIC Yellow Report.” Nucl. Phys.A 1026 (2022) 122447.

11. X. Ai et al. “A Common Tracking Software Project.” Computing and Software for Big Science 6, 8 (2022).

12. **J.D. Osborn** et al. “Implementation of ACTS into sPHENIX track reconstruction.” Computing and Software for Big Science 5, 23 (2021).

13. U. A. Acharya et al. (PHENIX Collaboration), “Probing gluon spin-momentum correlations in transversely polarized protons through midrapidity isolated direct photons in $p^{\uparrow} + p$ collisions at $\sqrt{s} = 200$ GeV. Phys. Rev. Lett. 127, 162001 (2021).

14. U. A. Acharya et al. (PHENIX Collaboration), “Transverse single-spin asymmetries of midrapidity π^0 and η mesons in polarized p+p collisions at $\sqrt{s} = 200$ GeV.” Phys. Rev. D 103, 052009 (2021).

15. C. A. Aidala et al. “Design and beam test results for the 2D projective sPHENIX electromagnetic calorimeter prototype.” IEEE Transactions on Nuclear Science, vol. 68, no. 2, pp. 173-181, Feb. 2021.

16. J. D. Osborn for the sPHENIX Collaboration, “Requirements, status, and plans for track reconstruction at the sPHENIX experiment.” arXiv:2007.00771, peer reviewed Connecting the Dots 2020 Workshop proceedings.

17. U. Acharya et al. (PHENIX Collaboration), “Measurement of jet-medium interactions via direct photon-hadron correlations in Au+Au and d+Au collisions at $\sqrt{s_{NN}} = 200$ GeV.” Phys. Rev. C 102, 054910 (2020).

18. J. D. Osborn for the LHCb Collaboration, “Jet hadronization at LHCb.” PoS High-p_T 2019 (2020) 013.

19. I. Helenius, J. Lajoie, **J. D. Osborn**, P. Paakkinen, H. Paukkunen, “Nuclear gluons at RHIC in a multi-observable approach.” Phys. Rev. D 100, 014004 (2019).

20. A. Aaij et al. (LHCb Collaboration), “Measurement of charged-hadron production in Z-tagged jets in proton-proton collisions at $\sqrt{s} = 8$ TeV.” Phys. Rev. Lett. 123, 232001 (2019).

21. C. Aidala et al. (PHENIX Collaboration), “Nonperturbative transverse momentum broadening in dihadron angular correlations in $\sqrt{s_{NN}}$= 200 GeV proton-nucleus collisions.” Phys. Rev. C 99, 044912 (2019).

22. J. D. Osborn for the PHENIX Collaboration, “PHENIX results on jet modification with π^0- and photon-triggered two particle correlations in p+p, $p(d)$+Au, and Au+Au collisions.” Nuclear Physics A 982, 591-594 (2019).

23. C. Aidala et al. (PHENIX Collaboration), "Nonperturbative transverse momentum dependent effects in dihadron and direct photon-hadron angular correlations in p+p collisions at $\sqrt{s} = 200$ GeV." Phys. Rev. D 98, 072004 (2018).

24. C. Aidala et al. (PHENIX Collaboration), "Single-spin asymmetry of J/ψ production in p+p, p+Al, and p+Au collisions with transversely polarized proton beams at $\sqrt{s_{NN}} = 200$ GeV." Phys. Rev. D 98, 012006 (2018).

25. J. D. Osborn for the PHENIX Collaboration, "Study of cold and hot nuclear matter effects on jets with direct photon triggered correlations from PHENIX." Nuclear Physics A 967, 476-479 (2017).

26. A. Adare et al. (PHENIX Collaboration), "Nonperturbative-transverse-momentum effects and evolution in dihadron and direct photon-hadron angular correlations in p+p collisions at $\sqrt{s} = 510$ GeV." Phys. Rev. D95, 072002 (2017).

Conference and Workshop Organization

1. Computing for High Energy Physics, Bangkok, Thailand, May 25-29, 2026.
2. Acts for NP Workshop, Berkeley, California, May 12-15 2025.
3. sPHENIX Commissioning Workfest, Brookhaven National Laboratory, March 13-17 2023.
4. sPHENIX Commissioning Workfest, Brookhaven National Laboratory, December 15-16 2022.
5. Second AI4EIC Workshop, College of William & Mary, October 10-14, 2022.
6. Acts Developers Workshop, CERN, September 26-30, 2022.
7. Streaming Readout 9 Workshop, virtual, December 8-10, 2021.
8. 2021 CFNS Summer School Software Lectures, virtual, August 9-23, 2021.
9. ECCE Second Simulation Workshop, virtual, May 21, 2021.
10. ECCE Simulation Workshop, virtual, April 2, 2021.
11. LHCP Conference QCD Organizing Committee, Paris, France, June 7-12, 2021.
12. Spin and EIC workshops at the RHIC All Users Meeting, Brookhaven National Laboratory, June 2-5, 2019.
13. American Physical Society Division of Particles and Fields Meeting, Ann Arbor, MI, August 3-7, 2015.

Collaboration and Research Community Service

DUNE PHLEX Review committee, 2025
Served as an external reviewer for the DUNE offline computing framework towards CDR approval.

HSF Trigger and Reco Convener, 2024-2026
Served as one of the conveners for the trigger and reconstruction software group in the HEP Software Foundation.

sPHENIX Analysis Coordinator, 2024-present
Served as sPHENIX analysis coordinator, overseeing development of all data processing activities towards physics analysis.

TPC Review Committee, October 20, 2023
Served on review committee for TPC repair review.

Chaired ePIC Backwards RICH Review, March 20-21, 2023
Chaired ePIC internal review with external committee members to provide recommendations for backwards RICH selection in ePIC baseline design.

ePIC Global Detector and Integration Convener, 2022-2023
Served as convener of the global detector and integration working group for the EPIC collaboration.

ePIC Computing and Software Convener, 2022-2023
Served as convener of the simulation and QA working group for the EPIC collaboration..

JINST, NIMA Reviewer 2021-present
Serve as a peer reviewer for articles submitted to the Journal of Instrumentation (JINST)

ECCE Software and Computing Convener, 2021-2022
Served as convener of the software and computing working group for the ECCE proto-collaboration.

sPHENIX Publication Policy Committee, 2021-2022
Served on committee developing sPHENIX publication policies and criteria.

sPHENIX Track Reconstruction Convener 2020-present
Served as convener of the track reconstruction software effort for the sPHENIX collaboration.

sPHENIX TPOT Reviewer, 12/2020
Review committee for TPC Outer Tracker for sPHENIX.

DOE Office of Nuclear Physics SBIR Proposal Reviewer, 11/2020
Review SBIR proposals related to computing in Nuclear Physics.

Nuclear Physics Day on Capitol Hill, 04/2018.
Met with staff members of Michigan Senators and Representatves to discuss funding for nuclear physics research.

PHENIX Executive Council, 01/2016-01/2017.
Served as elected member to executive council, which advises PHENIX spokesperson on scientific priorities.

Invited Conference and Workshop Presentations

1. **Plenary**: Fast Machine Learning for Science Conference, August 31-September 4, 2026. " A Foundation Model for Nuclear and Particle Physics."
2. Acts Developers Workshop, April 20-24, 2026. "Acts in sPHENIX."
3. Second Workshop on Trajectory Reconstruction in Particle Physics, July 21-23, 2025. "ACTS Integration into the sPHENIX Experiment."
4. ACTS4NP, May 12-15, 2025. "Alignment within Acts."
5. Quark Matter 2025, April 6-10, 2024. "Λ_c/D^0 ratio measurement at sPHENIX."
6. Acts Developers Workshop, November 18-21, 2024. "Acts at sPHENIX."
7. 2nd CFNS Workshop on Energy Flow, November 6-9, 2023. "sPHENIX calorimeter highlights and physics."
8. Acts Developers Workshop, November 7-10, 2023. "Acts in sPHENIX."
9. RHIC/AGS All Users Meeting, June 7-10, 2022. "sPHENIX Spin and Cold QCD Future Physics Program."
10. Connecting the Dots Workshop, May 31-June 2, 2022. "4D track reconstruction with sPHENIX."
11. Streaming Readout 10 Workshop, May 17-19, 2022. "4D track reconstruction with sPHENIX."
12. Probing QCD at High Energy and Density with Jets INT Workshop, August 16-20, 2021. "Recent results and future directions for jet hadronization and substructure measurements."
13. CFNS EIC Summer School, August 9-23, 2021. "ECCE software stack and detector geometry."
14. EICUG Summer Meeting, August 2-6, 2021. "ECCE software."
15. RHIC Science Program Toward EIC in the Coming Years, May 24-26, 2021. "Photon-jet and dijet probes at RHIC towards the EIC."
16. **Plenary**: Computing in High Energy Physics (CHEP) Conference, May 17-21, 2021. "Implementation of ACTS into sPHENIX Track Reconstruction."
17. Streaming Readout VIII Workshop, April 28-30, 2021. "ORNL Streaming Readout Plans at the EIC."
18. Resummation, Evolution, Factorization Workshop, December 7-11, 2020. "Experimental perspective on hadronization."
19. Jets for 3D Imaging Workshop, November 23-25, 2020. "Jet substructure at the EIC."
20. ACTS Workshop, May 25-29, 2020. "sPHENIX Experience with Acts."
21. Second EIC Yellow Report Workshop at Pavia University, May 20-22, 2020. "Inclusive and heavy flavor jet substructure at the EIC."

22. Connecting The Dots Workshop, April 22-24, 2020. "Requirements, status and plans for track reconstruction of the sPHENIX experiment."
23. 3^{rd} JETSCAPE Workshop, March 18-20, 2020. "Hadronization and jet substructure at RHIC and the LHC."
24. 13^{th} International Workshop on High p_T Physics in the RHIC/LHC Era, March 19-22, 2019. "Jet hadronization at LHCb."
25. Workshop on Novel Probes of the Nucleon Structure in SIDIS, e+e- and pp (FF2019), March 14-16, 2019. "Factorization breaking, color entanglement, and hadronization of jets."
26. Workshop on the Definition of Jets, Brookhaven National Laboratory, June 25-27, 2018. "Probing effects from QCD color with photon-jet and dijet correlations."
27. Quark Matter 2018, Lido, Italy, May 13-19, 2018. "PHENIX results on jet modification with π^0- and photon-triggered two particle correlations in p+p, p(d)+Au, and Au(Cu)+Au collisions."
28. Quark Matter 2017, Chicago, Illinois, February 5-11, 2017. "Study of cold and hot nuclear matter effects on jets with direct photon-triggered correlations from PHENIX."
29. CMS SMP-J Workshop Annual Workshop, January 25, 2017. "Color entanglement and color coherence."

Seminars and Colloquia

1. BNL Nuclear Physics Seminar, September 16, 2024. "Foundation Models for Nuclear and Particle Physics."
2. Stony Brook EIC Summer School lectures, June 3-14, 2024. "Tracking and Calorimetery."
3. University of Tennessee seminar, March 15, 2022. "sPHENIX charged particle reconstruction."
4. Oak Ridge National Laboratory seminar, June 16, 2021. "Addressing data reduction challenges with next generation scientific software."
5. Workflow Workshop Seminar Series, August 14, 2020. "Data processing workflows at scattering user facilities."
6. University of Tennessee HEP/Nuclear/Astro Seminar, February 3, 2020. "Jet substructure at RHIC and the LHC."
7. University of Kentucky HEP/Nuclear Seminar, November 7, 2019. "Hadronization and jet substructure at RHIC and the LHC."
8. University of Michigan HEP/Astro/Nuclear Seminar, April 8, 2019. "Jet substructure at RHIC and the LHC."
9. Oak Ridge National Laboratory Seminar, February 18, 2019. "Peering inside protons and nuclei."

10. Wayne State University PAN Seminar, September 14, 2018. "Effects from color flow in proton-proton and proton-nucleus collisions."
11. Paul Laurence Dunbar High School Senior Seminar, August 31, 2018. "Career paths with a physics degree."
12. University of Illinois HEP/MEP Colloquium, October 23, 2017. "Effects from color entanglement in proton-proton and proton-nucleus collisions."
13. University of Michigan HEP/Astro/Nuclear Seminar, November 21, 2016. "Recent experimental results on QCD factorization breaking at RHIC."
14. Brookhaven National Laboratory Nuclear Seminar Series, October 25, 2016. "Recent experimental results on QCD factorization breaking at RHIC."
15. Seminar at Columbia University, October 24, 2016. "Recent experimental results on QCD factorization breaking at RHIC."